

Kinetic Theory of Gases

SHORT NOTES

- ❖ **Boyle's law:** If m and T are constant

$$V \propto \frac{1}{P}$$

$$P_1 V_1 = P_2 V_2$$

- ❖ **Charles's law:** If m and P are constant

$$V \propto T$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- ❖ **Gay-Lussac's law:** If m and V are constant

$$P \propto T$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- ❖ **Avogadro's law:** If P , V and T are same

$$N_1 = N_2$$

where, N_1 and N_2 are the number of molecules

$$V \propto n \text{ (no. of molecules of gas)}$$

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- ❖ **Graham's law:** If P and T are constant

$$\text{rate of diffusion } r \propto \frac{1}{\sqrt{\rho}}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

- ❖ **Dalton's law:** $P = P_1 + P_2 + P_3 \dots$

P = Total pressure

$P_1, P_2, P_3 \dots$ = Pressure exerted by each component present in the mixture.

- ❖ **Ideal gas equation:** $PV = nRT = k_B NT$

n = number of moles

N = number of molecules

R = universal gas constant

k_B = Boltzmann's constant

- ❖ Pressure exerted by ideal gas

$$P = \frac{1}{3} \frac{mN}{\text{Volume}} \overline{v^2}$$

$$\overline{v^2} = \text{mean square velocity} \left\{ \overline{v^2} = \frac{v_1^2 + v_2^2 + v_3^2 + \dots}{N} \right\}$$

m = mass of each molecule

$$\text{or } P = \frac{1}{3} nm \overline{v^2}$$

$$n = \text{number density i.e., } n = \frac{N}{\text{Volume}}$$

$$\text{❖ } V_{\text{rms}} = (\overline{v^2})^{1/2}$$

$$\text{K.E.} = \frac{3}{2} k_B T = \frac{1}{2} m \overline{v^2}$$

$$\overline{v^2} = \frac{3k_B T}{m}$$

$$\Rightarrow v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$$

$$\text{❖ } v_{\text{av}} = \sqrt{\frac{8k_B T}{\pi m}}$$

$$\text{❖ } v_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}$$

$$\text{❖ Mean free path } (\overline{l}) = \frac{1}{\sqrt{2} n \pi d^2}$$

n = number density

d = diameter of molecule

$$\text{❖ } \gamma = \frac{C_p}{C_v}$$

γ = ratio of specific heats

C_p = specific heat at constant pressure

C_v = specific heat at constant volume

$$C_p - C_v = R$$

R = universal gas constant

S.No.	Atomicity	No. of degree of freedom	C_p	C_v	$\gamma = C_p/C_v$
1	Monoatomic	3	$\frac{5}{2}R$	$\frac{3}{2}R$	$\frac{5}{3}$
2	Diatomic	5	$\frac{7}{2}R$	$\frac{5}{2}R$	$\frac{7}{5}$

$$\text{❖ } C_{v(\text{mix})} = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$$

where n_1 and n_2 are number of moles of gases mixed together
 C_{v_1} and C_{v_2} are molar specific heat at constant volume of the two gas and $C_{v(\text{mix})}$: Molar specific heat at constant volume for mixture.